

《数据库系统原理》实验报告（）					
题目：数据库实验					
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实验环境： 硬件环境： windows\wsl, Docker Desktop 版本 28.0.1 软件环境： OceanBase 社区版镜像:oceanbase/oceanbase-ce:lastest					
实验步骤及结果截图： 1. 建立数据库 Major，在 Major 数据库中创建表 Players <pre>obclient(root@sys)[(none)]> CREATE DATABASE Major; Query OK, 1 row affected (0.033 sec) obclient(root@sys)[(none)]> USE Major; Database changed obclient(root@sys)[Major]> CREATE TABLE Players (-> Sno VARCHAR(10) PRIMARY KEY, -> Sname VARCHAR(20) UNIQUE, -> Ssex VARCHAR(1), -> Sage SMALLINT, -> Steam VARCHAR(20) ->); Query OK, 0 rows affected (0.052 sec)</pre> 2. 插入表单项 <pre>obclient(root@sys)[Major]> INSERT INTO Players (Sno, Sname, Ssex, Sage, Steam) -> VALUES -> ('200215121', 'LuLu', 'M', 27, 'G2'), -> ('200215122', 'TiMo', 'M', 18, 'G2'); Query OK, 2 rows affected (0.021 sec) Records: 2 Duplicates: 0 Warnings: 0</pre> 3.建立用户 Jay <pre>obclient(root@sys)[mamba]> Create User 'Jay' -> Identified BY '2351136'; Query OK, 0 rows affected (0.020 sec) obclient(root@sys)[mamba]> GRANT CREATE ,GRANT OPTION ON *.* TO 'Jay'; Query OK, 0 rows affected (0.018 sec)</pre> 4. 在 Jay 用户下尝试查询 Players。 登录 Jay 用户 <pre>obclient(root@sys)[mamba]> exit Bye sh-4.4# obclient -uJay -p -h127.1 -P2881 Enter password: Welcome to the OceanBase. Commands end with ; or \g. Your OceanBase connection id is 3221487645 Server version: OceanBase_CE 4.2.1.10 (r110000072024112010-28c1343085627e79a4f13c29121646bb889cf901) (Built Nov 20 2024 10:11:18) Copyright (c) 2000, 2018, OceanBase and/or its affiliates. All rights reserved. Type 'help;' or '\h' for help. Type '\c' to clear the current input statement. obclient(Jay@[(none)][(none)]> []</pre>					

查询 Players

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
obclient(Jay@(none))[(none)]> SELECT * FROM Major.Players;
ERROR 1142 (42000): SELECT command denied to user 'Jay'@'%' for table 'Players'
[172.17.0.2:2882] [2025-04-22 09:37:17.046182] [YB42AC110002-00063359ACA077B4-0-0]
```

5. 授予用户 Jay 对表 Players 的查询、插入、修改等对象特权，并赋予其再授权的能力

```
obclient(root@sys)[(none)]> GRANT SELECT, INSERT, UPDATE ON Major.Players TO 'Jay' WITH GRANT OPTION;
Query OK, 0 rows affected (0.033 sec)
```

6. 在 Jay 用户下查询 Players（注意使用 Major.Players）

```
obclient(Jay@(none))[(none)]> SELECT * FROM Major.Players;
+-----+-----+-----+-----+-----+
| Sno      | Sname | Ssex | Sage | Steam |
+-----+-----+-----+-----+-----+
| 200215121 | LuLu  | M    | 27   | G2     |
| 200215122 | TiMo  | M    | 18   | G2     |
+-----+-----+-----+-----+-----+
2 rows in set (0.011 sec)
```

7. 在 Jay 用户下再插入一个元组的数据('200215123', 'ZywOo', 'M', 23, 'Vit'), 并在 Jay 用户下查询（注意使用 Major.Players）

```
obclient(Jay@(none))[(none)]> INSERT INTO Major.Players (Sno, Sname, Ssex, Sage, Steam)
-> VALUES ('200215123', 'ZywOo', 'M', 23, 'Vit');
Query OK, 1 row affected (0.004 sec)
```

```
obclient(Jay@(none))[(none)]> SELECT * FROM Major.Players;
+-----+-----+-----+-----+-----+
| Sno      | Sname | Ssex | Sage | Steam |
+-----+-----+-----+-----+-----+
| 200215121 | LuLu  | M    | 27   | G2     |
| 200215122 | TiMo  | M    | 18   | G2     |
| 200215123 | ZywOo | M    | 23   | Vit    |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

8. 删除 Jay 用户

```
obclient(Jay@(none))[(none)]> exit
Bye
sh-4.4# obclient -uroot@sys -h127.1 -P2881
Welcome to the OceanBase. Commands end with ; or \g.
Your OceanBase connection id is 3221487651
Server version: OceanBase_CE 4.2.1.10 (r110000072024112010-28c1343085627e79a4f13c29121646bb889cf901) (Built Nov 20 2024 10:11:18)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

obclient(root@sys)[(none)]> drop user 'Jay';
Query OK, 0 rows affected (0.024 sec)
```

出现的问题：

1. 授权命令报错

```
obclient(root@sys)[mamba]> CREATE USER 'Jay'@'localhost' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.020 sec)

obclient(root@sys)[mamba]> GRANT CREATE TABLE ON Major.* TO 'Jay'@'localhost' WITH GRANT OPTION;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase version for the right syntax to use near 'TABLE ON Maj
or.* TO 'Jay'@'localhost' WITH GRANT OPTION' at line 1
[172.17.0.2:2882] [2025-04-22 09:31:20.833277] [VB42AC110002-00063359AD60A502-0-0]
```

解决方案：

1. 查找了相关知识点和指令，发现并没有 **create table** 这一个权限，而是用 **create**，就能授予用户创建表格的能力

```
obclient(root@sys)[mamba]> Create User 'Jay'
-> Identified BY '2351136';
Query OK, 0 rows affected (0.020 sec)

obclient(root@sys)[mamba]> GRANT CREATE ,GRANT OPTION ON *.* TO 'Jay';
Query OK, 0 rows affected (0.018 sec)
```